CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Quick revision, one page

Cell basics

- The cell is the basic structural and functional unit of living organisms.
- Robert Hooke (1665) named cells in cork; Leeuwenhoek observed living microorganisms.
- Unicellular: one cell (amoeba, bacteria). Multicellular: many cells (humans, trees).

Parts of a cell and functions

- Cell membrane: selectively permeable, controls entry and exit.
- Cytoplasm: jelly-like fluid where cell activities take place.
- Nucleus: controls cell activities; holds genetic material.
- Cell wall (plants, fungi, bacteria): rigid, gives shape and protection.
- Chloroplast (plants): contains chlorophyll, site of photosynthesis.
- Vacuole: stores water and dissolved substances (large and central in plants).

Plant cell vs animal cell

- Plant: cell wall, chloroplasts, large central vacuole, fixed shape.
- Animal: no cell wall or chloroplasts, small vacuoles, irregular shape.

Microscope and sizes

- Total magnification $=$ magnification of eyepiece $\times$ magnification of objective.
- 1 millimetre $=1000$ micrometres.
- Prokaryotes (bacteria): no true nucleus; eukaryotes: true nucleus.
- Slide preparation: thin peel, stain (iodine/safranin), cover slip.

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